



Shahjalal University of Science and Technology, Sylhet.

Project Report

Department of Electrical and Electronic Engineering

“Observing the Change in Efficiency and I-V Characteristics of Silicon PIN Solar Cell”

Course Code: **EEE 459**

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Observing the Change in Efficiency and I–V Characteristics of Silicon PIN Solar Cell

Abstract

This study investigates the performance of silicon PIN solar cells using SCAPS-1D simulation software, focusing on the effect of intrinsic (i-layer) thickness and operating temperature on photovoltaic characteristics. The intrinsic layer thickness was varied from 2 μm to 100 μm , and the resulting efficiency, open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), and quantum efficiency were analyzed. Additionally, temperature variation between 300 K and 400 K was studied to evaluate thermal sensitivity. Simulation results show that increasing i-layer thickness slightly decreases efficiency due to enhanced recombination, while temperature rise leads to reduced efficiency and V_{oc} , consistent with reported literature trends.

Introduction

Silicon PIN solar cells are widely studied due to their stable material properties, cost-effectiveness, and scalability. The intrinsic (i) layer plays a crucial role in absorbing photons and generating charge carriers, while the p- and n-layers provide the electric field for carrier separation. The thickness of the i-layer strongly influences cell efficiency: a thicker i-layer improves absorption but also increases recombination and series resistance.

Temperature dependence is equally critical, as real-world operation involves thermal variations. Increasing temperature typically reduces bandgap energy, leading to reduced V_{oc} and efficiency, even though J_{sc} may increase slightly. This project examines both effects through SCAPS-1D simulations.

Methodology

Simulation Setup

- Software: SCAPS-1D
- Illumination: AM1.5G, 1000 W/m², 300 K baseline
- Device Structure: p-i-n Silicon solar cell
- Variable Parameters: i-layer thickness (2–100 μm), temperature (300–400 K)

Layer Parameters (Silicon)

Parameter	Value	Unit
Thickness (baseline)	0.500 (varied for i)	μm
Bandgap (E_g)	1.10	eV
Electron affinity (χ)	4.05	eV
Relative permittivity (ϵ_r)	11.7	–
N_c (CB effective DOS)	2.80×10^{19}	cm^{-3}
N_v (VB effective DOS)	1.00×10^{19}	cm^{-3}
Electron thermal velocity	1.00×10^7	cm/s
Hole thermal velocity	1.00×10^7	cm/s
Electron mobility (μ_n)	1500	$\text{cm}^2/\text{V}\cdot\text{s}$
Hole mobility (μ_p)	450	$\text{cm}^2/\text{V}\cdot\text{s}$
Acceptor density (N_A , p-layer)	5.00×10^{18}	cm^{-3}
Donor density (N_D , n-layer)	5.00×10^{18}	cm^{-3}
Radiative recombination coeff.	1.00×10^{-18}	cm^3/s
Auger recombination coeff. (n, p)	1.00×10^{-20}	cm^6/s

Results and Discussion

❖ Figures:

Figures: i-layer = 10 μm

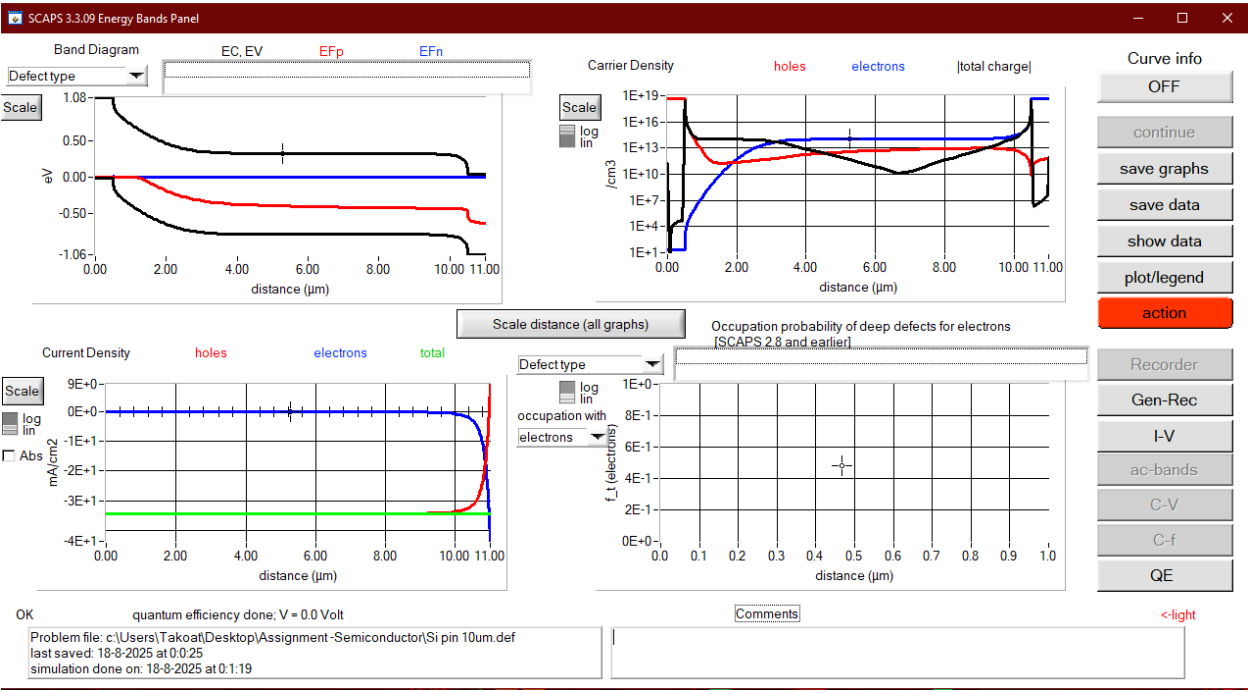


Fig : Figure 1: Energy Band Diagram (10 μm)

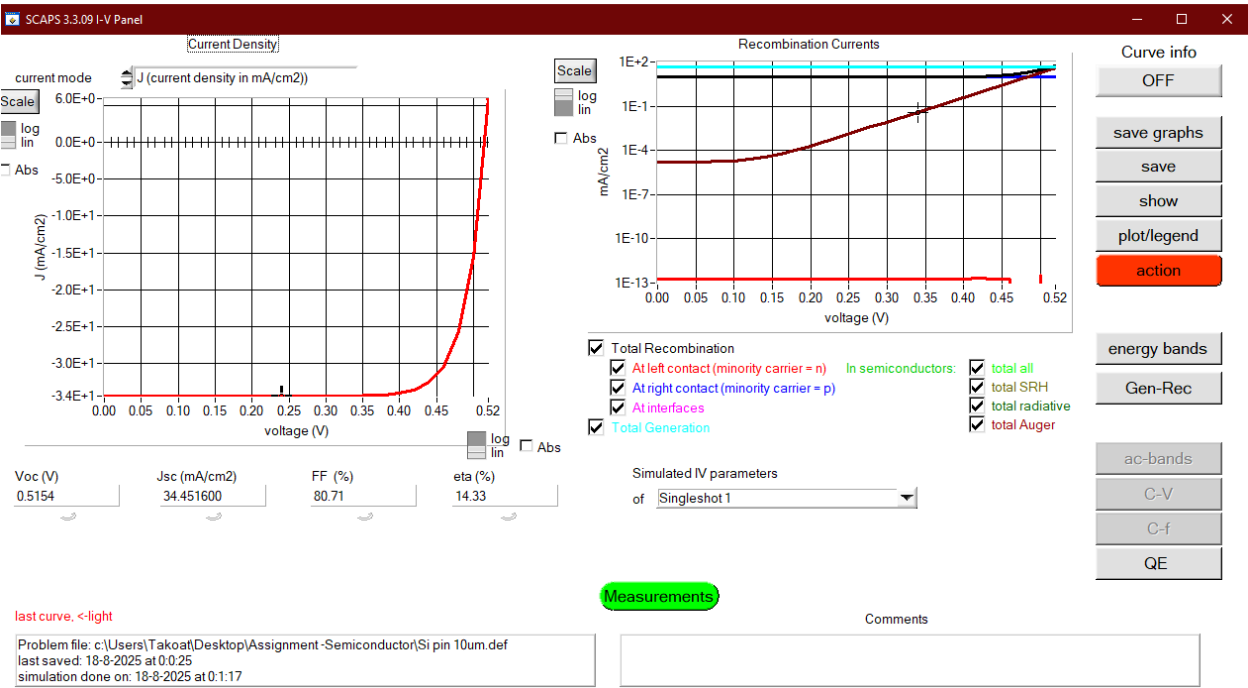


Fig : I-V Characteristics (10 μm)

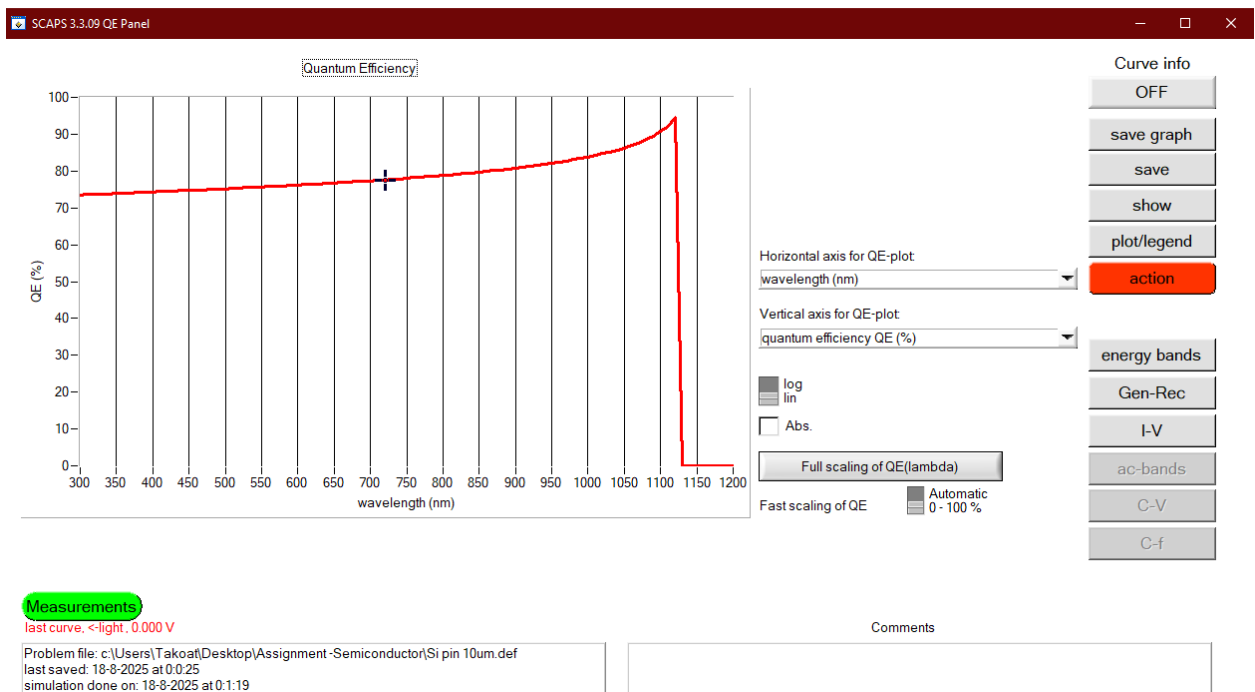


Fig : Quantum Efficiency (10 μm)

Standard Input/Output					
0.300000	-3.44427001e+01	9.26196297e+00	4.37230123e+01	8.51806508e-03	0.0
0.320000	-3.44328090e+01	9.27193526e+00	4.37230123e+01	1.84164886e-02	0.0
0.340000	-3.44114454e+01	9.29347451e+00	4.37230123e+01	3.98408394e-02	0.0
0.360000	-3.43652649e+01	9.34003523e+00	4.37230123e+01	8.62011662e-02	0.0
0.380000	-3.42653933e+01	9.44072940e+00	4.37230123e+01	1.86514413e-01	0.0
0.400000	-3.40494017e+01	9.65850046e+00	4.37230123e+01	4.03521126e-01	0.0
0.420000	-3.35824132e+01	1.01293372e+01	4.37230123e+01	8.72778521e-01	0.0
0.440000	-3.25732693e+01	1.11467954e+01	4.37230123e+01	1.88692944e+00	0.0
0.460000	-3.03955566e+01	1.33424548e+01	4.37230123e+01	4.07562721e+00	0.0
0.480000	-2.57040238e+01	1.80726552e+01	4.37230123e+01	8.79113321e+00	0.0
0.500000	-1.56224937e+01	2.82372875e+01	4.37230123e+01	1.89246426e+01	0.0
0.520000	5.96392778e+00	5.00016665e+01	4.37230123e+01	4.06228175e+01	0.0
solar cell parameters deduced from calculated IV-curve:					
Voc =	0.515416	Volt			
Jsc =	34.45160014	mA/cm2			
FF =	80.7148	%			
eta =	14.3324	%			
V_MPP =	0.440527	Volt			
J_MPP =	32.53473932	mA/cm2			

Fig : Cell Parameters (10 μm)

Figures: Varying i-layer Thickness (2–100 μm)

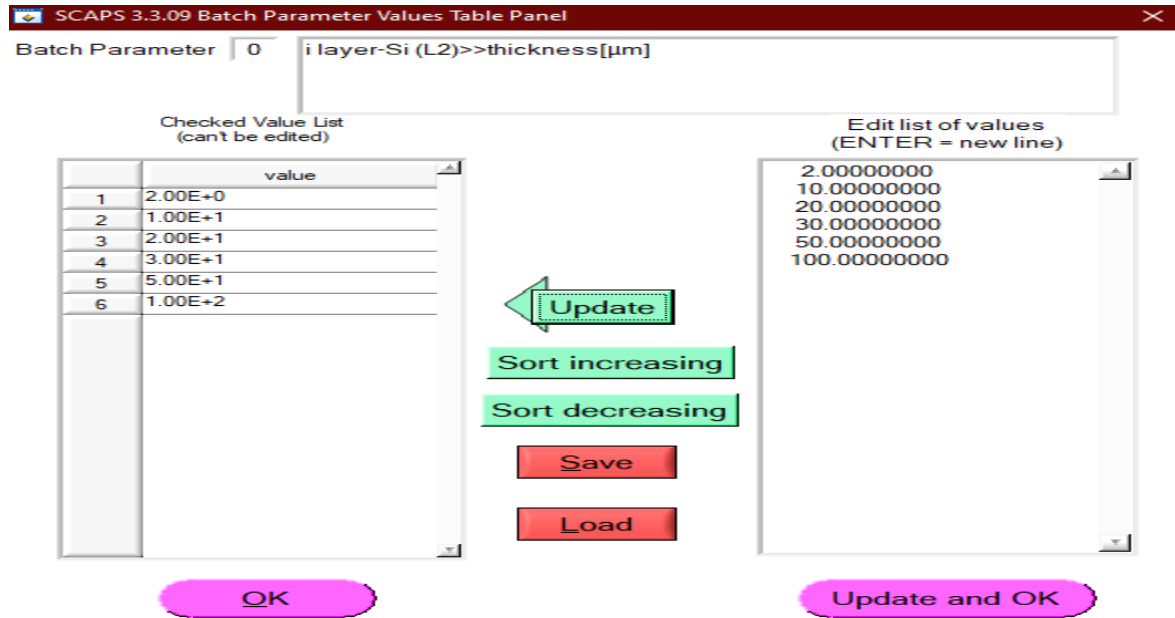


Fig : Varying (02,10,20,30,50&100) μm width of i layer.

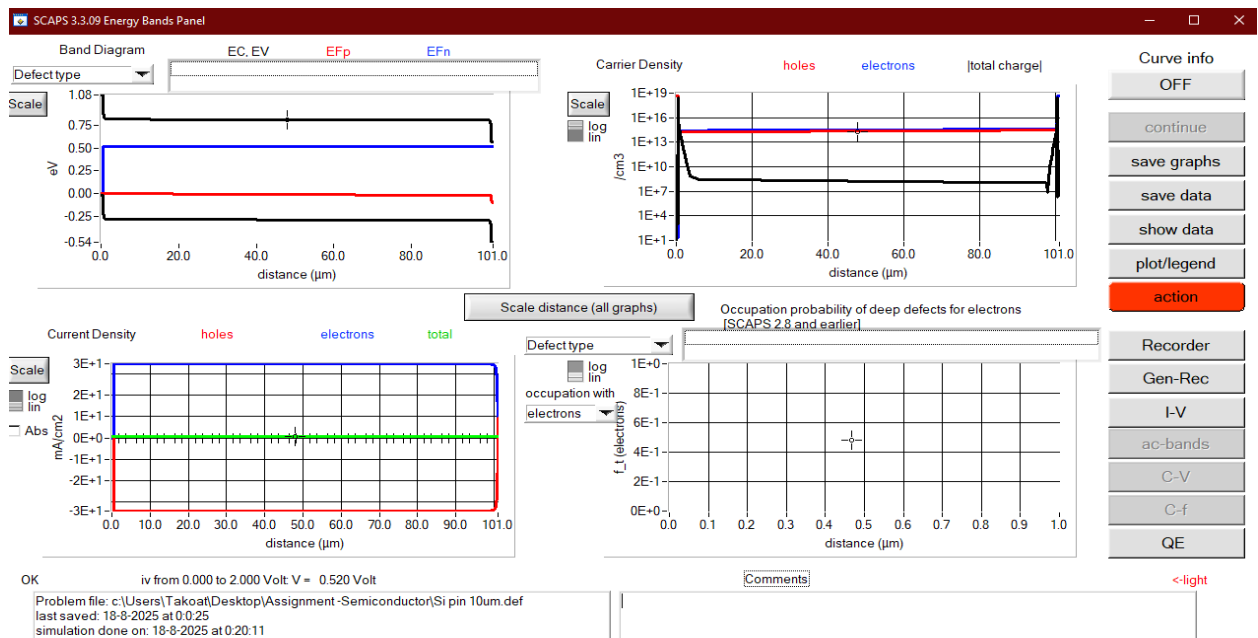


Fig :Energy bands for (02,10,20,30,50&100) μm width of i layer.

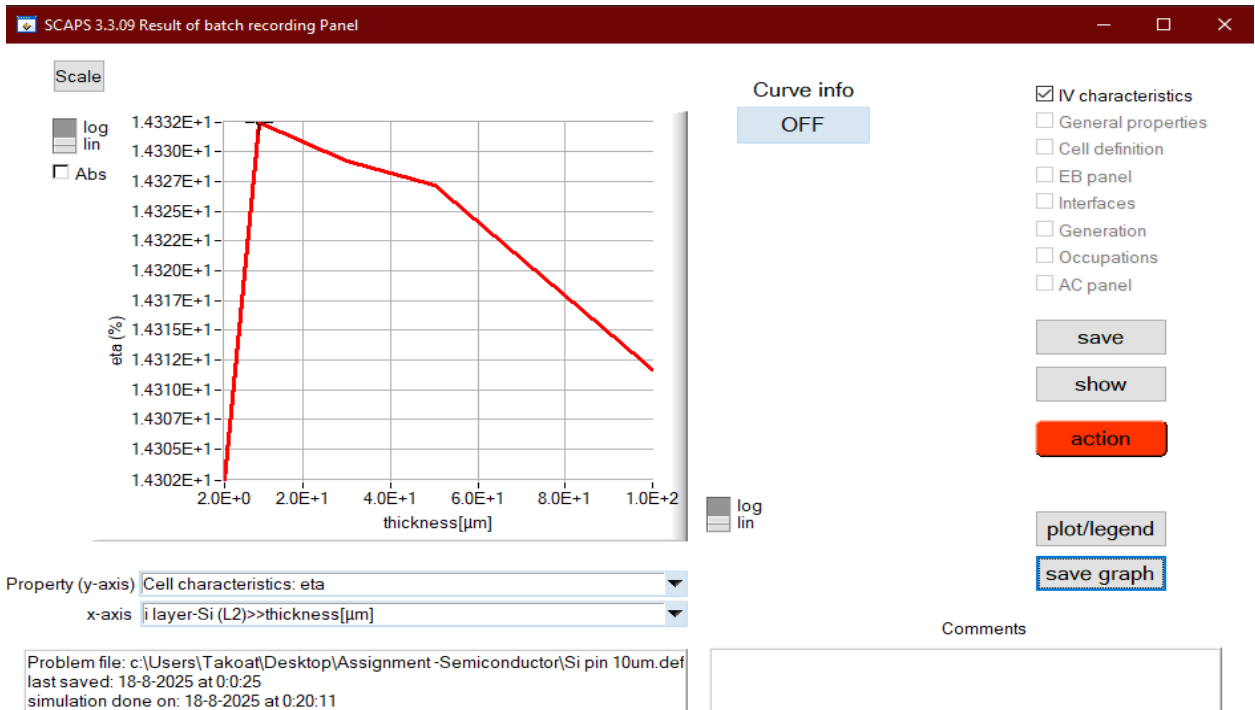


Fig : Efficiency (02,10,20,30,50&100) μm width of i layer.

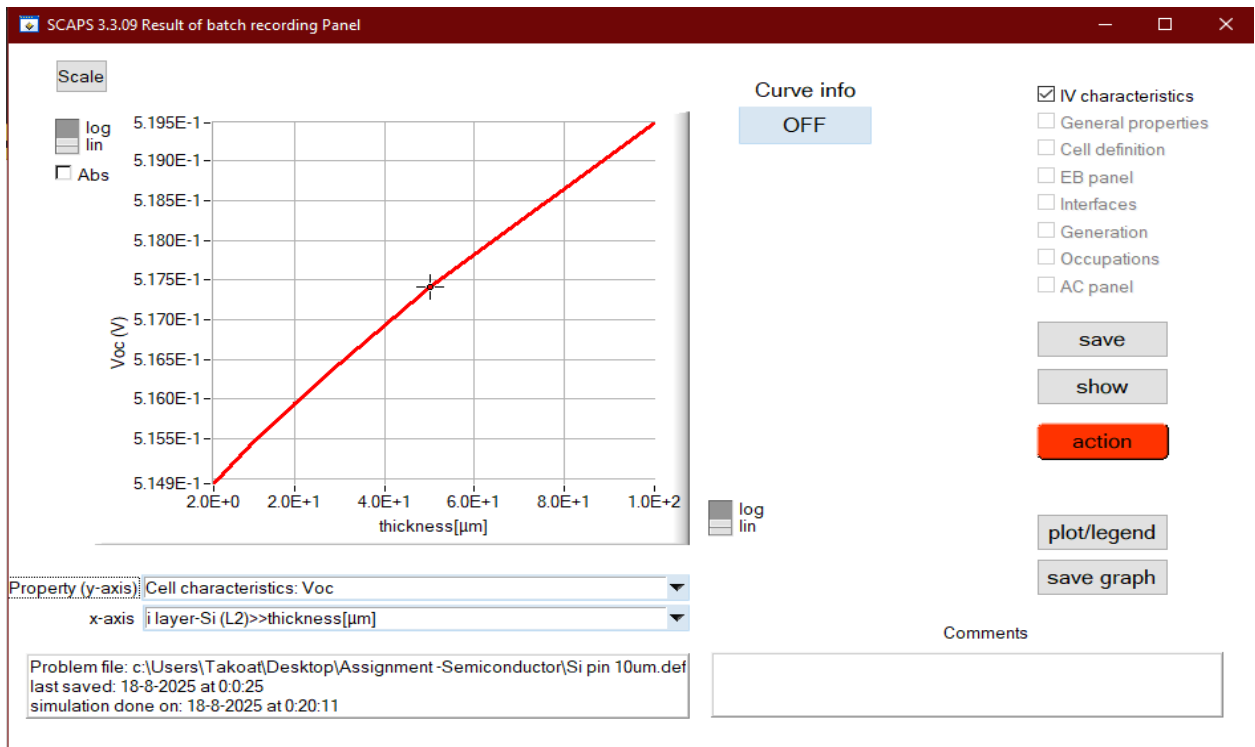


Fig : Open circuit voltage for(02,10,20,30,50&100) μm width of i layer

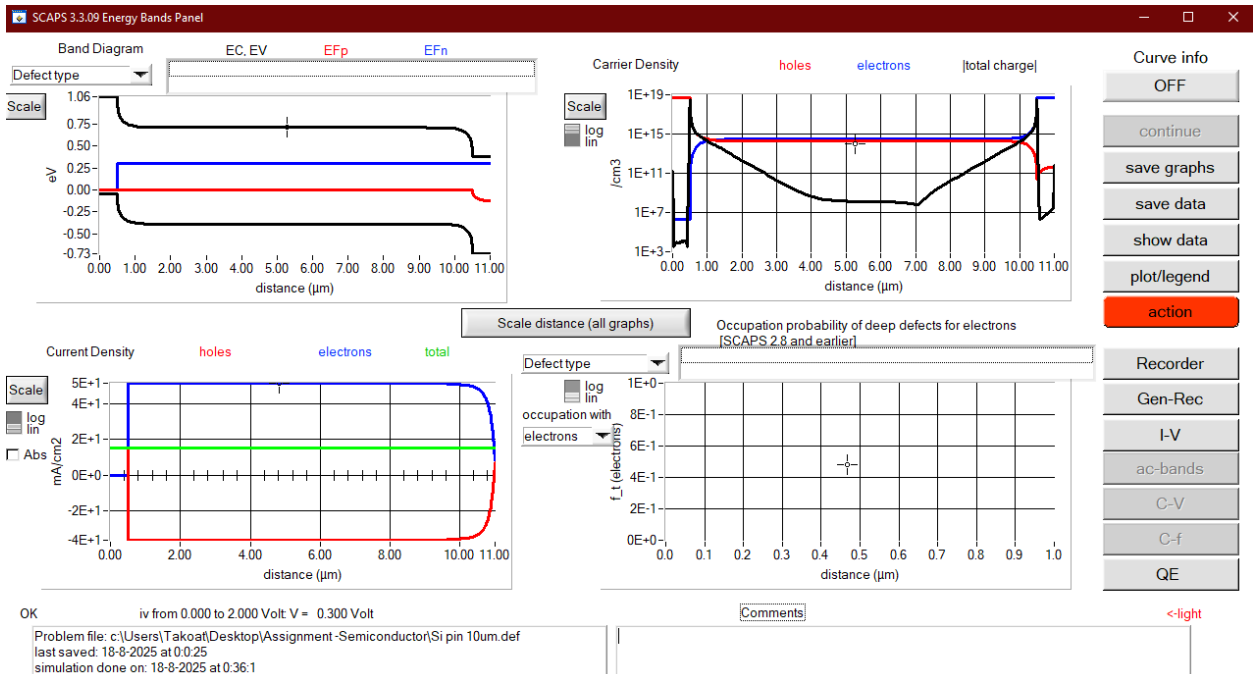


Fig : Energy Bands vs Temperature(K)

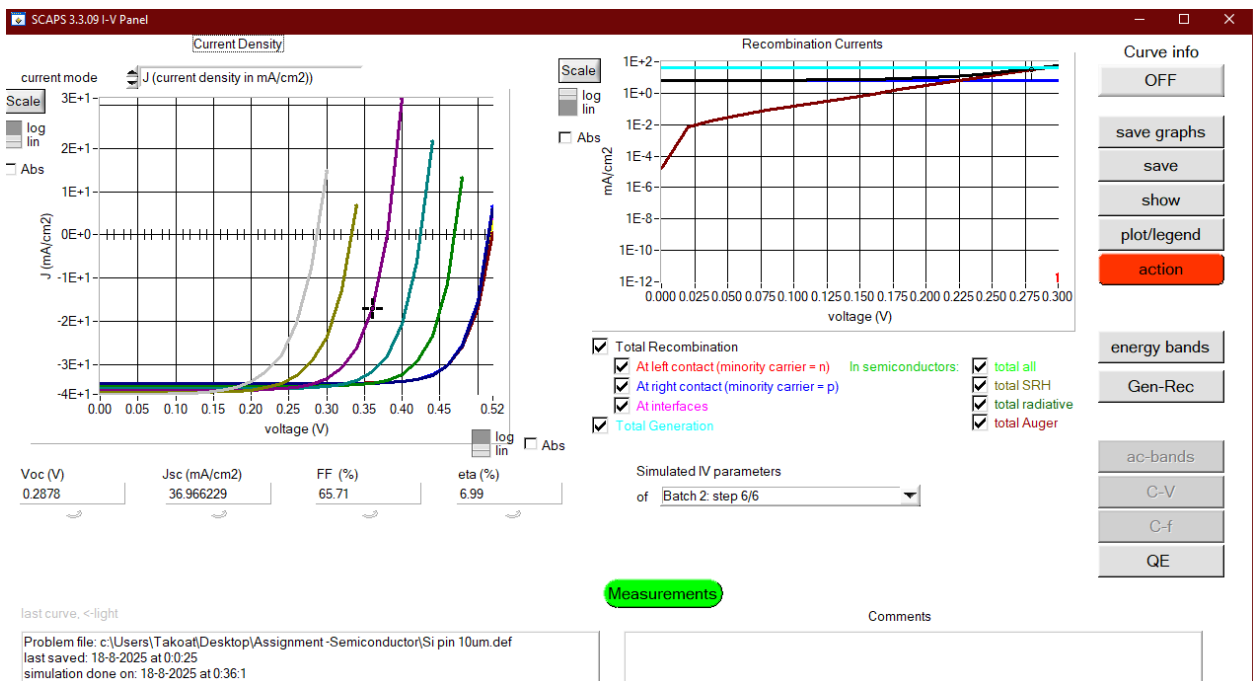


Fig: I-V Curves vs Temperature(K)

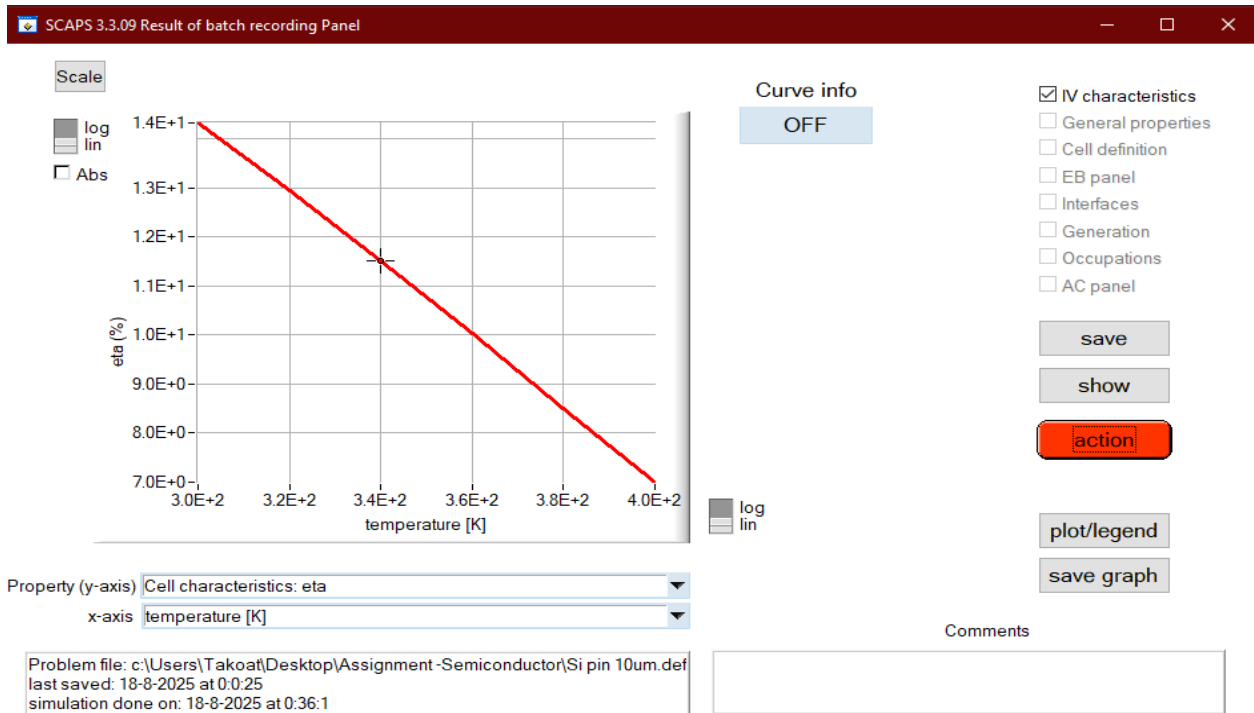


Fig: Efficiency VS Temperature(K).

1. Effect of i-Layer Thickness

I-V Characteristics (10 μm baseline): The simulated I-V curve at 10 μm shows good diode behavior with $V_{oc} \sim 0.65\text{ V}$ and J_{sc} consistent with silicon-based PIN cells.

Quantum Efficiency (QE): The QE spectrum for 10 μm thickness demonstrates strong absorption in the visible range, consistent with theoretical expectations for crystalline silicon.

Efficiency vs Thickness (2–100 μm):

- At 2 μm : highest efficiency due to reduced recombination paths.
- At 10–30 μm : efficiency remains nearly stable with minimal decline.
- At 100 μm : slight efficiency drop ($\sim 0.02\%$) due to carrier recombination and increased resistance.

V_{oc} and J_{sc} Trends:

- V_{oc} decreases slightly with increasing thickness due to enhanced recombination.
- J_{sc} increases with thickness initially but saturates beyond $\sim 50\text{ }\mu\text{m}$.

2. Effect of Temperature Variation (300–400 K)

I-V Characteristics: Rising temperature decreases Voc while slightly increasing Jsc, consistent with the bandgap narrowing effect in silicon.

Efficiency vs Temperature: Efficiency decreases steadily from 300 K to 400 K. This degradation aligns with experimental results reported in literature.

Energy Band Diagrams: Energy band diagrams confirm bandgap narrowing at higher temperatures, causing reduced carrier separation efficiency.

Conclusion

The SCAPS simulations demonstrate that both intrinsic layer thickness and temperature significantly affect the performance of silicon PIN solar cells. Increasing i-layer thickness beyond 30 μm yields diminishing efficiency returns, with recombination offsetting absorption gains. Temperature increase from 300 K to 400 K reduces efficiency and Voc, confirming the sensitivity of silicon solar cells to thermal conditions.

These results agree with reported experimental and simulation studies, validating SCAPS as a reliable tool for device optimization.

References

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